IT4010 PROGRESS PRESENTATION 2

GoodRoad: Citizen-Powered Traffic Violation Reporting System

ID: 25-26J-265 | Automating Enforcement with AI

Wijerathna P. G. S. P.
Deepfake Detection (MesoNet)

Rathnayake W. K. D.
Hybrid ANPR (YOLOv11)

Liyanage R. L. N. L.
Vision-Language Framework

Hendawitharana H. W. K. P.
Penalty & Reward Module

SLIIT | FACULTY OF COMPUTING

The Problem: An Invisible Enforcement Gap

Global road safety crisis meets lack of monitoring infrastructure

1.3 Million
Annual road deaths globally

87%
Involve traffic violations

99%
Road networks remain **UNMONITORED**

PRIMARY STAKEHOLDERS

Law Enforcement

Sri Lanka Police / Dept. of Motor Traffic (RMV). Beneficiaries of automated evidence collection.

SECONDARY STAKEHOLDERS

Citizens & Insurance

Public reporters acting as sensors and insurance companies verifying accident claims.

RESEARCH GAP

Existing systems lack the ability to verify **video authenticity (Deepfakes)** and lack **semantic reasoning** for complex violations.

GoodRoad: The Quad-Module AI Framework

A decentralized **Citizen-as-Sensor** network utilizing AI to automate evidence validation.

MesoNet

01

VIDEO AUTHENTICITY

Specialized deepfake detection and mesoscopic artifact screening to ensure video evidence integrity.

YOLOv11 + OCR

02

LOCALIZATION & RECOGNITION

Robust license plate detection and character recognition optimized for Sri Lankan vehicle plates.

Kimi-k2.5 VLM

03

SEMANTIC REASONING

Vision-Language Model providing semantic reasoning for complex traffic violations like "illegal turns".

Penalty & Reward

04

DETERMINISTIC LOGIC

Automated financial integrity module enforcing a strict 60/25/15 revenue split for all fines collected.

End-to-End System Architecture

5-Stage Processing Pipeline

01

Ingestion

React Frontend
FastAPI Backend

02

Security

Deepfake Detection

MesoNet
Deepfake Screening

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03

Localization

Hybrid ANPR System

YOLOv11 + SRGAN
OCR Engine

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04

Reasoning

Vision-Language Framework

Kimi-k2.5 VLM
Context Analysis

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05

Enforcement

Penalty & Reward Module

Email Alert
Reward Trigger

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Seamless data transfer via **MongoDB** persistence layer — Python AI Engines integrated throughout

Module 1 — Deepfake Detection

Wijerathna P. G. S. P. | Completion: 95% ✅

5-STAGE ANALYSIS PIPELINE

Upload → Frame Extraction → Artifact Screening → Temporal Consistency → Report Generation

KEY ACHIEVEMENTS

MesoNet Implementation

Deployed specialized CNNs to detect mesoscopic artifacts invisible to the human eye.

High Accuracy Benchmarks

Achieved >90% detection accuracy on FaceForensics++ and DeepFake-TIMIT datasets.

End-to-End Integration DONE

Full pipeline functional from video upload to JSON forensic report generation.

95%
COMPLETED

Module 2 — Vision-Language Framework

Liyanage R. L. N. L. | Active Status

Major Model Upgrade

Transitioned to **Kimi-k2.5** for multi-class traffic violation detection and detailed scene analysis.

KEY FEATURES

Semantic Reasoning

Explains **WHY** a maneuver is illegal, moving beyond simple bounding box detection.

HazardQA Dataset

Instruction-tuned on specialized dataset to understand complex traffic scenarios.

Human-Like Analysis

Simulates a traffic officer's reasoning process for ambiguous situations.

> *"Kimi-k2.5 enables the system to reason about complex violations beyond simple object detection."*

Module 3 — Hybrid ANPR System

Rathnayake W. K. D. | Completion: 92% ✅

HYBRID ANPR FLOW

Input Stream → SRGAN Upscale → YOLOv11 Detect → OCR Extract → Data Log

TECHNICAL HIGHLIGHTS

YOLOv11 Architecture

Superior plate localization accuracy and speed compared to prior YOLO versions.

SRGAN Enhancement

Super-Resolution upscaling restores readable details from low-res dashcam frames.

Key Innovation

Transition to YOLOv11 improved bounding box precision on Sri Lankan plate formats.

Module 4 — Penalty & Reward Module

Hendawitharana H. W. K. P. | Active Status

AUTOMATED REVENUE LOGIC

- Gov Tax **60%**
- Reward **25%**
- Sys Fee **15%**

Module Completion

Financial Logic & Integration

95%

Google Gemini Integration

Generates context-aware, educational violation notices to improve driver behavior.

MongoDB Schema

Finalized profiles for Vehicles, Violations, and Reward Logs for secure record keeping.

Deterministic Logic

Ensures unalterable financial integrity for every transaction processed.

Market Opportunity & Projected Impact

Commercialization Strategy

Value Proposition

Zero-CapEx

Infrastructure-free model for the government. Converts the entire road network into a monitored enforcement zone without installing new cameras.

Revenue Model

15% Cut

Sustainable revenue stream generated via a 15% commission from every successfully validated traffic violation fine.

Projected Impact

70% Drop

Projected reduction in traffic violations within the first 6 months of deployment due to the psychological "always-watched" effect.

GoodRoad transforms citizens into a distributed enforcement network — no camera infrastructure required.

Thank You